



Macroinvertebrate taxa richness uncertainty and kick sampling in the establishment of Mediterranean rivers ecological status

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ABSTRACT

Freshwater macroinvertebrates have been extensively used as environmental indicators and are the most prevalent biological group used in aquatic bioassessment in the European Water Framework Directive (WFD 2000/60/EEC), usually through several popular indices, as the Biological Monitoring Working Party (BMWP). Many of these indices are based on taxa richness, i.e. the number of taxa present in a given area, as the simplest and most common measure of biodiversity. Given the importance to the WFD of the ecological status assessment by macroinvertebrates and the consequences thereof, sampling requires careful consideration and evaluation of the associated uncertainty. In this work, carried out in a Mediterranean river, we show that after 20 sample “kicks” it was possible to estimate the true taxa richness using Clench nonlinear asymptotic models (CM). However, cumulative curves of taxa extracted with kick sampling underestimated the true number of theoretical taxa (A). In order to achieve an acceptable error a very large sample size was required, always >20 kicks. According to these criteria, sampling was clearly inefficient in most localities. The minimum effort required to achieve a significant and acceptable level of taxa richness, for 90% of A, should be between 25 and 71 kicks and for 95% of A, 52–150 kicks. Both satisfactory percentages represent a mean difference of 2 (range 0–6) and 3 (range 1–8) taxa actually not being captured from the total predicted for each locality, for 90 and 95% of the CM asymptote, respectively. This study shows that by using the 20 kicks methodology it is possible to achieve reliable true macroinvertebrate richness estimates, but the establishment of the community composition, i.e. the full taxa making up any index score, will be inaccurate to an unknown degree.

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1. Introduction

The implementation of the European Water Framework Directive (WFD 2000/60/EEC; EC, 2000) requires the use of aquatic organisms or biological quality elements (BQE) to assess the ecological status of bodies of water. These BQEs include phytoplankton, aquatic flora, macroinvertebrates and fish. Among these, the freshwater macroinvertebrates have been extensively used as environmental stress bioindicators of inland water bodies for more than 100 years and are the most prevalent biological group used in aquatic bioassessment in the WFD (Rosenberg and Resh, 1993; Birk et al., 2012). Despite this long history, there have been relatively few studies to establish macroinvertebrate sampling protocols to obtain reliable field data. Birk et al. (2012) asked whether “the effort to acquire the biological data is sufficient and feasible to fulfil the WFD assessment objectives.” As sampling greatly influences the results of bioassessment, macroinvertebrate collection must be precise and representative. At the same time, sampling has to be cost-effective (Vlek et al., 2006).

Given the importance to the WFD of the ecological status assessment by macroinvertebrates and the consequences thereof, sampling requires careful consideration and evaluation of the associated uncertainty (Carter and Resh, 2001). Without this assessment, assigning

water quality status according to the WFD is extremely problematic. This is especially so in the case of indices which depend on taxa richness as the number of taxa found depends on the sampling effort (Gotelli and Colwell, 2001). Several popular indices based, directly or indirectly, on taxa richness are among the WFD indicators of ecological status, including the Biological Monitoring Working Party (BMWP) (Armitage et al., 1983) and all its derivatives. Taxa richness, defined as the number of taxa present in a given area, is the simplest and most common measure of biodiversity (Magurran, 2004). But despite their simplicity, taxonomic richness show sample size dependency, and have been shown to underestimate richness at all sample sizes, although the error diminishes as the sample size increases (Hellmann and Fowler, 1999; Schreiber and Brauns, 2010). Nevertheless, this drawback is rarely considered in biomonitoring.

The determination of the ecological status of water masses usually relies on the macroinvertebrate community composition rather than on taxa richness. On the other hand, the qualitative composition found will depend on the sampling effort. Thus, to measure the ecological status in an objective scientifically rigorous manner using an index derived from the community, as the aforementioned BMWP, it is essential to take enough samples to reach a minimally acceptable objective of community richness. Without taking into account the effect of sample size on measures of taxa richness, the bioassessment results are likely to yield false quality assignments that will be translated into inadequate, even harmful, management practices. It is important, then, that the methodology to determine the true value of the taxa present in a certain location should be assessed in terms of its

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uncertainty, including bias, precision and accuracy of measurement (Walther and Moore, 2005) (Fig. 1). Since the uncertainty depends on data measurement, conclusions could be compromised by the error committed in data gathering in terms of times, places, specialists, methodologies, etc. Determining the true value is not a simple task, especially when the number of taxa found in a community increases with the sampling effort (Baltanás, 1992; Melo and Froehlich, 2001; Magurran, 2004).

One of the most commonly used methods to collect freshwater macroinvertebrates is the semi-quantitative ‘kick-sampling’ (Macan, 1958), the standard procedure used in many biological quality indices (e.g. Armitage et al., 1983; Alba-Tercedor and Sánchez-Ortega, 1988; Capítulo et al., 2001; Mustow, 2002; Czerniawska-Kusza, 2005; among others). The application of this technique must be reliable in terms of its reproducibility, repeatability and accuracy. That is, the method must be able to produce the same results in a temporally and spatially homogeneous local community. To do so demands precise calibration and the definition of measurement errors, so as to refine the precision of the technique and to optimize the sampling protocols for local conditions (Bradey and Ormerod, 2002; Clarke et al., 2002; Vlek et al., 2006; Feeley et al., 2012). At the same time, methods need to be developed to correct for the effect of sampling effort on taxa richness found (Walther and Martin, 2001).

Here we present taxa richness estimation methods that yield estimates of the true richness of a location. We follow two approaches: taxa accumulation curves (TACs) and non-linear asymptotic models, which offer increased reliability for inventories, the possibility of comparison and better sampling planning (Colwell and Coddington, 1994; Thompson et al., 2003; Magurran, 2004; Chao et al., 2005). The usual kick sampling mode, based on a few samples, provides an uncertain estimate of the true taxa richness in a location. But the use

of a standardized kick sampling protocol and the calculation of TACs, based on a consistent sample size, enables an estimation of the true taxa richness to be made.

The aims of this study were (i) to assess the performance of the “kick sampling” method using cumulative curves of observed macroinvertebrates taxa richness against five indicators: 1) estimated true richness through extrapolation, 2) percentage achieved with respect to the theoretical maximum [indicators 1) and 2) obtained from the total sampling effort after the calculation of nonlinear asymptotic models]; 3) bias, 4) precision and 5) accuracy; (ii) to discuss the feasibility/viability of a widely accepted sampling technique/protocol to obtain reliable inventories of the aquatic macroinvertebrate community.

2. Material and methods

2.1. Study location

The study was conducted on the river Múrtigas (Guadiana basin, SW Iberian Peninsula), a typical Mediterranean river, 81 km in length and with a 478 km² drainage basin. The River Múrtigas runs between 200 and 1000 m a.s.l. The area has a temperate continental Mediterranean climate (Martínez-Santos et al., 2014), with an average annual temperature between 14° and 16 °C and a mean rainfall between 800 mm and 1000 mm, concentrated in autumn and spring. To avoid bias due to temporal heterogeneity, sampling was carried out intensively between October and November 2010, in 15 localities distributed more or less equally along the initial 55 km stretch of the river (Fig. 2). This option may pose some limits to temporal inferences different from the period when the sampling was carried out, i.e. autumn.

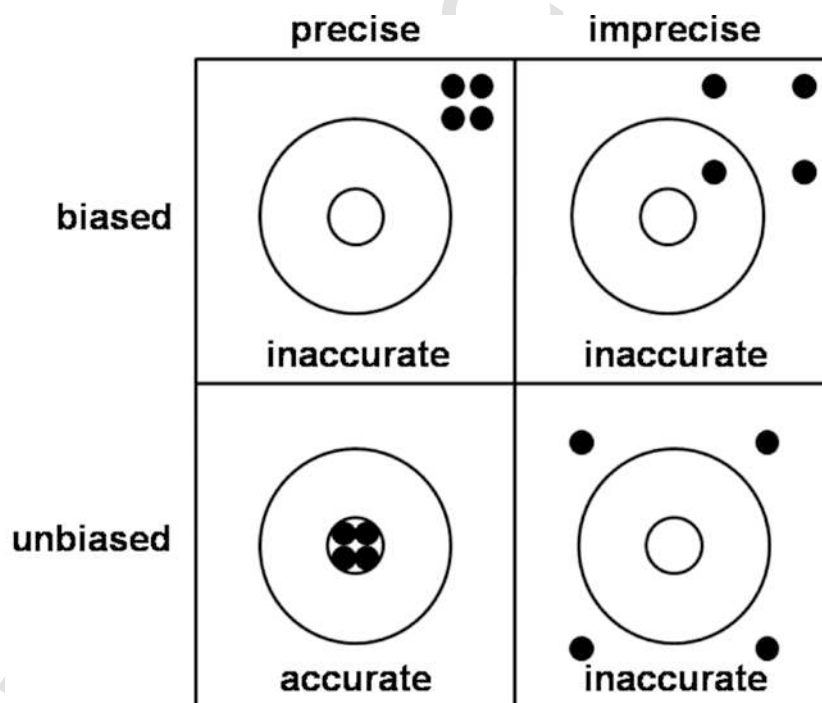


Fig. 1. Bias, precision and accuracy. Bias is the difference between the observed average richness and the accepted reference value or actual value, determining the under- or overestimates of the true value. Precision, or random error, is the variance of the observed values, independently of the accepted real value. Accuracy is defined as the total distance between the observed values and the accepted true value.

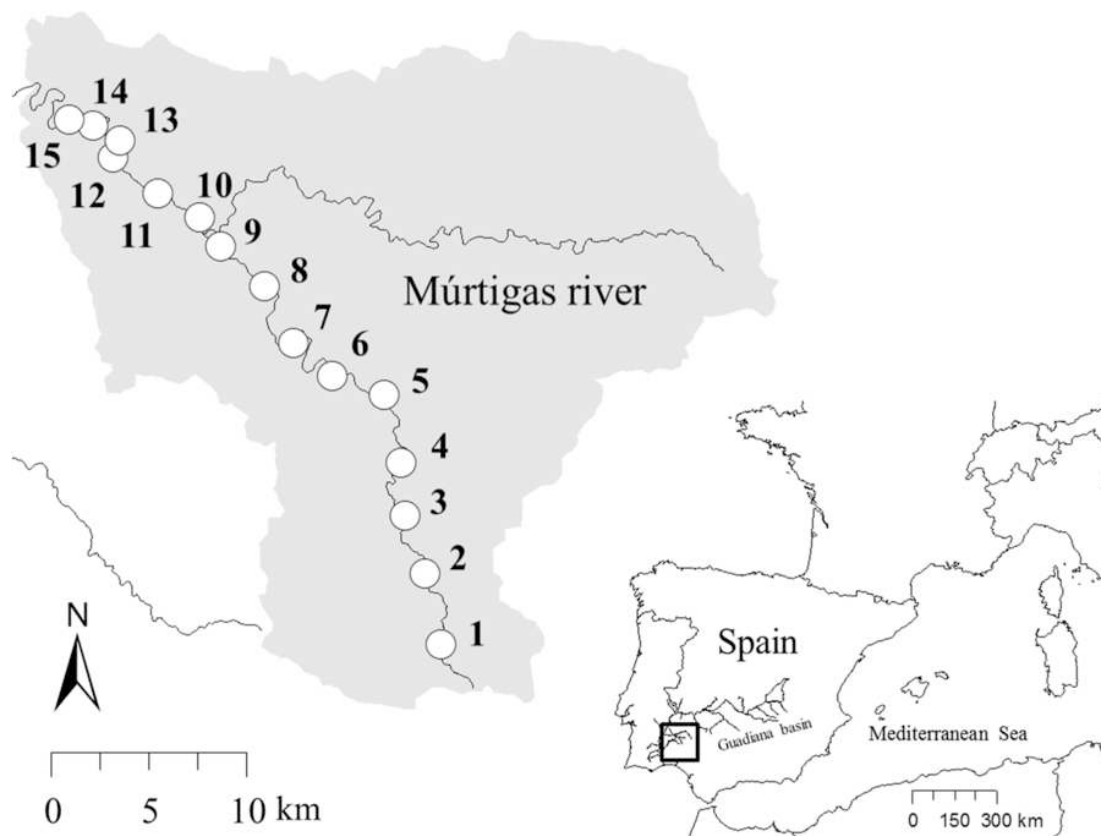


Fig. 2. Macroinvertebrates sampling localities (open circles) in the River Múrtigas (SW Spain). The Múrtigas basin is shown in grey.

2.2. Sampling

A sampling protocol was systematically applied so that surveys could provide a robust and unbiased image in terms of reproducibility. A 100-m river stretch was selected in each locality for macroinvertebrate sampling.

The standardized “kick sampling” applied was based on the “The Development and Testing of an Integrated Assessment System for the Ecological Quality of Streams and Rivers throughout Europe using Benthic Macroinvertebrates project” (AQEM project) (Hering et al., 2003). Macroinvertebrate sampling was carried out following the multihabitat semiquantitative approach (Barbour et al., 1999). After visually identifying the different mesohabitats (hard substrate, plant debris, vegetated banks, submerged macrophytes and sand and other fine sediments) present in each 100 m sampling stretch, a percentage of cover was visually assigned to each mesohabitat. This methodology involves a sampling unit, or kick, for every 5% coverage of each mesohabitat found in the 100 m stretch.

A quadrangular hand net, 0.25 m wide, 0.5 m deep and with a mesh size of 250 μ m, was used to collect the invertebrates. The sampling unit was one “kicking” that consisted of the agitation and re-suspension of the substrate 0.5 m from the hand net for 30 s (Cortés et al., 2002). The sample was cleaned *in situ* and stored in a labelled plastic bottle, using 70% ethanol as a preservative for subsequent laboratory analysis. In total, 20 kicks in each locality were sampled, which represent approximately a total of 5 m² of fluvial substrate. Before kick sampling, all taxa of surface macroinvertebrates, such as Gyrinidae, Gerridae or Hydrometridae, were recorded, as they are usually not captured with this procedure.

In the laboratory, the taxa were identified using different keys (Puig et al., 1984; Tachet et al., 2000), to family level (except oligochaetes, ostracods and Hydrachnidae) using a Nikon SMZ1000 stereomicroscope.

2.3. Data analysis

2.3.1. Richness estimates

At each location we calculated the observed taxa accumulation curve (TAC), through successive sampling effort units (up to 20 kicks), to determine the number of taxa extracted, or observed total richness “ S_{obs} ” (Moreno and Halfpeter, 2000). To avoid any possible influence by the order in which the kicks were taken in the field, a randomization procedure was applied to eliminate the random error (Colwell and Coddington, 1994; Thompson and Thompson, 2007) from 100 permutations applied to each of the observed curves using the EstimateS 8.20 software (Colwell, 2011). The result was a smoothed curve of accumulation for the observed taxa (e.g. Willott, 2001), by averaging 100 curves randomized for each locality (S_{est} , an estimate at each sample size, according to the “ S_{obs} Mao Tau” methodology for sample-based rarefaction curves without replacement). The Mao Tau analytic method provides a confidence interval at 95% for the current taxa present in each location (see formulas in Colwell et al., 2004, 2012).

When the TACs reach a horizontal asymptote, it is considered that the sampling effort is sufficient to determine the true richness of taxa, and consequently the inventory is complete (Soberón and Llorente, 1993). In cases where this asymptote is not achieved, fitting nonlinear asymptotic models allows us to estimate the true richness of taxa by

extrapolation to a hypothetically infinite sampling effort (Moreno and Halfpeter, 2000; Canning-Clode et al., 2008).

Obtaining the true richness, including the taxa that have not been found during the study but are probably present, is a laborious task in the case of highly rich faunal groups (Willott, 2001; Hortal et al., 2006) as is the case of the macroinvertebrates in Mediterranean rivers (Bonada et al., 2007a). However, it is possible to find the asymptote of incomplete inventories and to make a comparison by extrapolation (Moreno and Halfpeter, 2001), through the use of non-linear asymptotic models (González-Oreja et al., 2010) such as the widely used *Clench model* (CM) (Clench, 1979). The CM was adjusted to the smoothed curves of taxa accumulation (S_{est}). For each location, the “estimated CM asymptote” (A) was considered the “true richness” and the terms are used interchangeably. Similarly the “estimated CM asymptote” was used as truth in order to assess the performance of kick sampling.

The CM assumes that the probability of adding new taxa decreases with the number of taxa already registered, but increases with the sampling effort (eq. (1)):

$$S_{clench}(t) = at / (1 + bt), \quad (1)$$

where t is a measure of the effort (in this case the number of kicks); $S(t)$ is the predicted number of taxa in effort t , or predicted richness; a represents the rate of increase at the beginning of sampling, and b is the accumulation of taxa, rate of change or curve shape.

The estimated asymptote was obtained according to the formula a/b . To calculate the percentage of A achieved we employ $S_{obs}/(a/b)$. For adjustments of the models to the smoothed curves, a non-linear regression was applied using *Simplex* and *Quasi-Newton* as the iterative method (e.g. Hortal et al., 2004). To assess the proportion of variance explained by the function and therefore the model that best fits each locality, the coefficient of determination (R^2) was used (Melo and Froehlich, 2001).

2.3.2. Bias, precision and accuracy

Bias, precision and accuracy (Fig. 1) were studied following Walther and Moore (2005). Bias (scaled mean error, *SME*, eq. (2)) is the average of all the differences between taxa richness predicted at each replica and the asymptote (A). This parameter reflects the degree by which A is under- or overestimated. In this case, the S_{obs} *Mao Tau* curve will always produce an underestimation of this value (González-Oreja et al., 2010; Băncilă et al., 2014), designated by a negative symbol. Precision was measured by the coefficient of variation (CV, eq. (3)), regardless of the A value, where SD means standard deviation. This parameter measures the repeatability of independent estimates. Ideally a TAC should achieve values of near-zero bias and high precision (Walther and Martin, 2001). Accuracy (scaled mean square error, *SMSE*, eq. (4)) takes into account the estimated value for each sampling effort and A, the true taxa richness predicted, where E_j is the taxa richness estimated in each unit effort for each j_{th} replica, n is the number of replicates per unit effort ($n = 100$ replicates), and $\bar{E}$ is the average of 100 estimates of richness observed in each unit of effort.

$$SME = \frac{1}{An} \sum_{j=1}^n (E_j - A), \quad (2)$$

$$CV = 100 \frac{SD}{\bar{E}}, \quad (3)$$

$$SMSE = \frac{1}{A^2 n} \sum_{j=1}^n (E_j - A)^2, \quad (4)$$

Equations were standardized to enable comparison of the outcomes for different communities and studies (Walther and Moore, 2005; Willie et al., 2012). According to Walther and Moore (2005), the evaluation of the development of an estimator or TAC should be performed when the estimates are still growing. To this end, calculations were performed for the average of all effort units and for five individual sampling efforts (3, 5, 10, 15 and 20 kicks).

Finally, since 100% of taxa requires an infinite effort (Soberón and Llorente, 1993), 90% and 95% of the total fauna was used as a conservative level of inventory integrity, enabling the comparison and estimation of the minimum effort required to achieve this level (see Moreno and Halfpeter, 2000). To reach these targets eq. (5) was applied to the Clench model, where q is the proportion of total fauna for which the sampling effort is desired (Jiménez-Valverde and Hortal, 2003).

$$tq_{clench} = q / (b * (1 - q)), \quad (5)$$

3. Results

3.1. Assessment of taxa accumulation curves (TACs)

A total of 67 taxa were captured through 300 kicks distributed across 15 locations (Appendix A). Richness observed per location ranged between 13 and 35 taxa. In all locations this parameter increased with sampling effort, but there were obvious differences between locations (Fig. 3).

The Clench model (CM) was fitted to the smoothed curves to predict A through extrapolation, with a very high coefficient of determination ($R^2 > 0.990$, except location 7). The asymptotes predicted by CM varied between 15 and 40 taxa (Table 1).

There were no curves that reached the asymptote or saturation level predicted by CM (Fig. 3). With the exception of location 9, none of the sites achieved 90% taxa capture, considered the lowest conservative level of inventory integrity. This shows that the sampling strategy was ineffective, with values of between 80 and 90% of taxa captured per site or less, as in locations 3 and 4. The asymptotes represent an average of 5 families above the maximum average S_{obs} per location (range: 2–10) (Table 1). These differences are relatively high if our aim is to determine the ecological value of a body of water.

Smoothed TACs were different from one location to another. Some curves had a slow growth at the start of sampling (e.g. location 3) while others grew faster (e.g. location 1) (Fig. 3). Furthermore, localities with equal richness displayed differences in the shape of the curves at the beginning (e.g. locations 12 and 13) and different rates of change or b values. In locations 4 and 15 asymptotes were further away, leaving the true richness much above the observed values (Table 1; Fig. 3). The difference between the observed and predicted richness was also highly variable across sites.

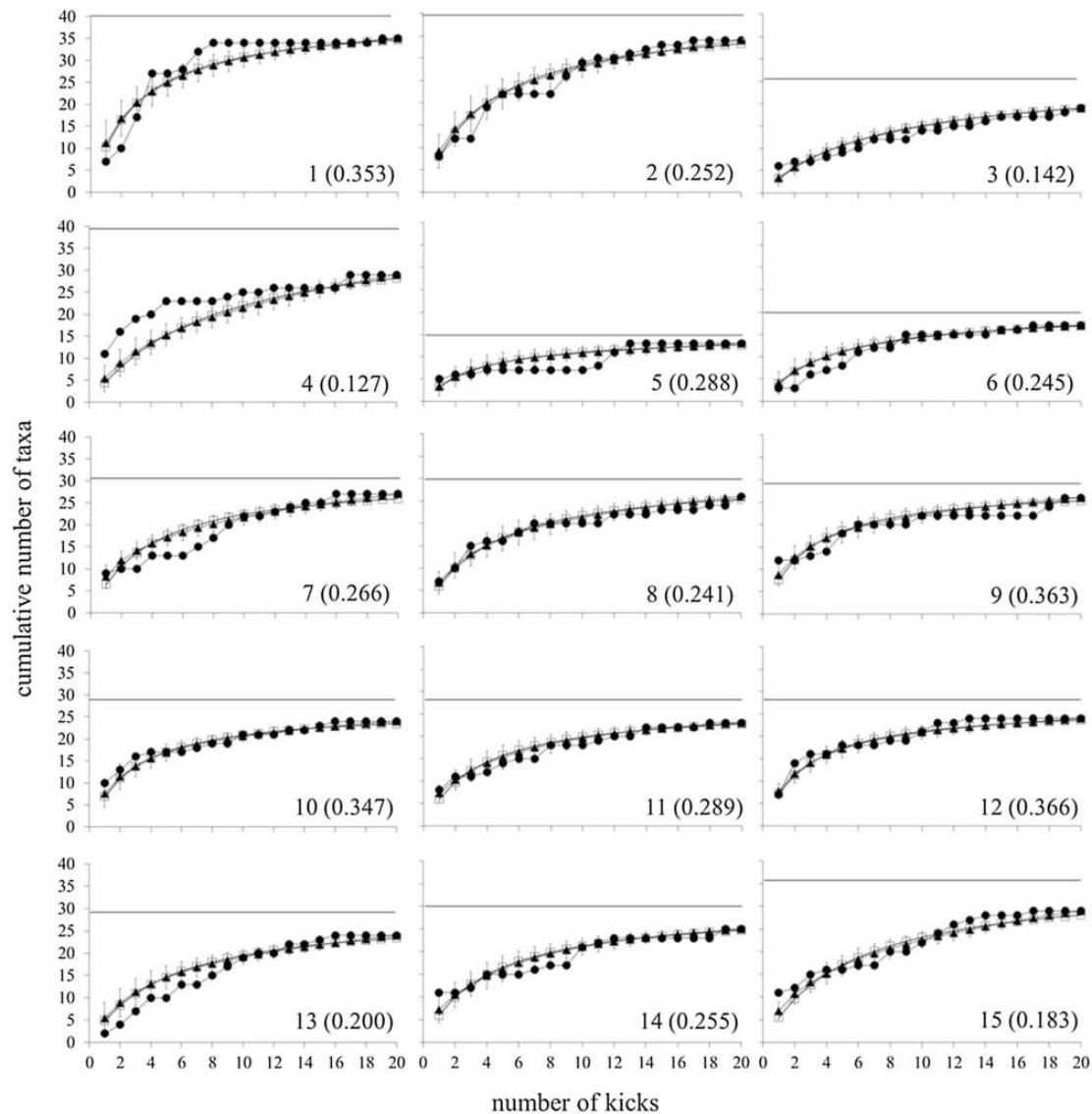


Fig. 3. Observed macroinvertebrate taxa accumulation curves and adjusted models for 15 localities sampled in the Múrtigas River (SW Spain). Dashed lines: observed data (S_{obs} ; filled circles); expected data (smoothed curves; filled triangles) $\pm$ error bars (standard deviation, SD). Solid line is the adjusted Clench model (CM, empty squares). Solid line and expected data are usually superimposed, because of the very high Clench adjustment. Horizontal line is the asymptote predicted by the CM. In brackets the rate of change (b), see Eq. (1).

3.2. Bias, precision and accuracy

The evaluation metrics employed to determine bias, precision and accuracy, i.e. SME, CV and SMSE, varied negatively with the sampling effort (Fig. 4) and also varied with location. At the beginning of the curve, the kick sampling showed high bias, low accuracy and low precision, which progressively diminished as the sampling effort increased (Fig. 4). In order to achieve an acceptable rate of error, e.g. a cut off point of 90% of the estimated CM asymptote (e.g. $SME > -0.1$, $CV < 5\%$ and $SMSE < 0.01$) a very large sample size is required, always >20 kicks. According to these criteria, sampling was clearly inefficient in all localities, except in site 9 where it approached efficiency (see Appendix B).

Cumulative curves of taxa extracted with kick sampling underestimated (negative bias) the true number of theoretical taxa (A) (Fig. 3). The CV always tends to zero as the effort increases (Fig. 4), which

maximizes precision. In Fig. 3, error bars (measured as the standard deviation) averaged over 100 randomizations, showed a clear reduction in amplitude as the sampling effort increased. Similarly, the amplitude of the error bars for accuracy and precision also markedly diminished as the sampling effort increased, but less in the case of bias (Fig. 4).

3.3. Determination of minimum sampling effort

An acceptable minimum sampling effort must fulfil two criteria: 1) the observed/expected taxa richness ratio (O/E) has to approach 1, and 2) the sampling error (bias, accuracy and precision) has to be minimized. According to Moreno and Halfiter (2000) the minimum threshold values for criteria 1 should be 90–95%. On the other hand the 90% O/E ratio cut-off point means $SME > -0.1$, $CV < 5\%$ and $SMSE < 0.01$.

Table 1

Results of the adjustment of the Clench model (CM) that describes the taxa accumulation curves with respect to the sampling effort of aquatic macroinvertebrates in 15 locations in the River Múrtigas (SW Spain). S_{obs} : observed richness; the 95% confidence interval refers to the number of taxa present as Mau Tau ($EstimateS$, see materials and methods). Clench Model. A : asymptote (a/b); a and b parameters; R^2 : coefficient of determination; %: percentage achieved relative to the asymptote [$S_{obs}/(a/b)$]. Sampling effort to reach an acceptable proportion of richness (kicks) is in the two last columns. $N = 20$ kicks in all cases.

Locations	S_{obs} 95%		Lower limit	Upper limit	Clench Model		A	R^2	%	Sampling effort to satisfactory richness	
	S_{obs}	95%			a	b				90%	95%
1	35	30	40	13.9516	0.3529	40	0.998	88.5	26	54	
2	34	29	39	10.0596	0.2526	40	0.995	85.4	36	75	
3	19	16	22	3.6451	0.1424	26	0.999	74.2	63	13	
4	29	23	35	4.9818	0.1266	39	0.994	73.7	71	15	
5	13	9	17	4.2656	0.2875	15	0.996	87.6	31	66	
6	17	15	19	4.9804	0.2449	20	0.999	83.6	37	78	
7	27	19	35	8.1884	0.2658	31	0.980	87.6	34	71	
8	26	22	30	7.4022	0.2413	31	0.995	84.7	37	75	
9	26	21	31	10.4513	0.3630	29	0.991	90.3	25	52	
10	24	20	28	9.3352	0.3472	27	0.995	89.3	26	55	
11	23	20	26	7.7017	0.2887	27	0.992	86.2	31	66	
12	24	18	30	9.8285	0.3664	27	0.998	89.5	25	52	
13	24	18	30	5.8506	0.2000	29	0.996	82.1	45	95	
14	25	22	28	7.5475	0.2553	30	0.995	84.6	35	74	
15	29	24	34	6.5566	0.1832	36	0.990	81.0	49	10	
Average	25	20	30			30	0.994	84.6			

The O/E taxa richness ratio for the different locations shows that, except location 9, none of the remaining 14 locations achieved a minimum satisfactory level >90% (Table 1). The sampling effort required to achieve a satisfactory level of inventory (90–95% of the CM asymptote or A), varied between different locations, and with the exception of location 9, it always fell short of efficient levels (Table 1). The minimum effort required to achieve a significant and acceptable level of taxa richness, for 90% of A , should be between 25 and 71 kicks and for 95% of A , 52–150 kicks (Table 1). Both satisfactory percentages represent a mean difference of 2 (range 0–6) and 3 (range 1–8) taxa actually not being captured from the total predicted for each locality, for 90 and 95% of the CM asymptote, respectively. The locations that need a greater sampling effort to achieve the asymptote or the proposed percentage are those whose slope is lower at the beginning and which are farther from the asymptote (high bias and less accuracy) (Table 1; Fig. 3).

4. Discussion

4.1. Performance of taxa accumulation curves with kicking method

The European Water Framework Directive (WFD 2000/60/EEC; EC, 2000) requires the determination of the ecological status of European water masses. This is normally done using biological quality indices that depend on taxonomic richness, like the very popular BMWP, (e.g. Birk and Hering, 2006). Thus, knowing the true richness on which these biological quality indices will depend is an essential objective in order to scientifically determine the exact environmental quality of any water mass. The consequences of an inadequate ecological status assignment as a result of inaccurately determining richness may cause public authorities to take inadequate or even harmful measures in the management and conservation of water masses (Armitage et al., 1983; Soberón and Llorente, 1993; Moreno and Halfpeter, 2000; Thompson et al., 2007). Inventories of highly diverse groups, such as aquatic macroinvertebrates, are difficult to

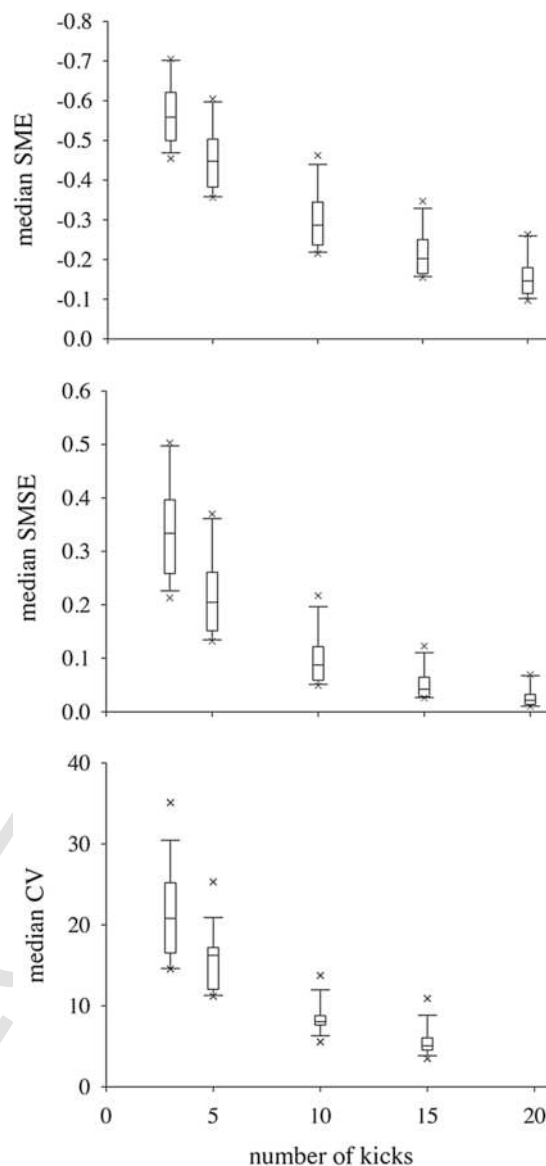


Fig. 4. Median box plot of scaled mean error (SME, bias), scaled mean squared error (SMSE; accuracy) and coefficient of variation (CV, precision) of macroinvertebrate total richness estimates from the Múrtigas river (SW Spain) for different sampling effort (kicks). Error bars: range; x: outlier values.

complete, as evidenced by the results presented here. We used taxa accumulation curves (TACs) as a scientifically reliable method that provides an estimate for true taxa richness (Colwell and Coddington, 1994; Thompson and Thompson, 2007; Thompson et al., 2007).

This study demonstrates the positive correlation between sampling effort and freshwater macroinvertebrate taxa richness from a Mediterranean river, as has previously been observed for many other plant and animal communities (Hellmann and Fowler 1999; Walther and Martin 2001; Gotelli and Colwell, 2001; Herzog et al., 2002). Also it shows that taxa richness, observed through a multihabitat kick sampling methodology incorporating 20 equivalent units of effort, underestimated the true richness of freshwater macroinvertebrate communities. Furthermore, the error of the estimates varied in different communities, regardless of sampling effort. As sample size increased, bias gradually decreased and accuracy and precision both increased, at different rates, depending on the localities. This was also

found in a study conducted by Vlek et al. (2006) on macroinvertebrates, which proposed that sample size should be based on a single specific mesohabitat within bioassessment works, instead of the multihabitat approach used here. However, the multihabitat approach provides a better representation of the communities found in a water mass and thus a more precise diagnostic of the general health of the ecosystem, as many others have previously observed (Karr et al., 1986; Carter and Resh, 2001; Hering et al., 2004).

There could be a certain limitation in the temporal applicability of these results to times other than autumn. But, the general tendencies should be coherent with them. As the patchiness in the macroinvertebrate distribution and/or taxa richness would change so to will the TACs. The differences will affect the asymptotes and the errors committed. However, in a permanent Mediterranean river as the Múrtigas, probably the optimal option to carry out a sample like this is in autumn, when flow is minimum and macroinvertebrate distribution is less patchy and taxa richness intermediate (Downes et al., 1993; Bonada et al., 2007b).

4.2. Assessment of asymptotic models performance

Soberón and Llorente (1993) determined that extrapolation through models enables the comparison of field data from butterfly and mammal inventories for conservation purposes. They employed Clench models (CM) (Clench, 1979), widely used to determine true richness (e.g. González-Oreja et al., 2010). In these works the goodness of fit of CM varied according to the features of the inventories. In a similar work to that presented here, Moreno and Halfpeter (2000) studied bat richness, setting a range of sampling effort through Clench and Linear Dependence models and also using extrapolation to determine the asymptote that represented the maximum theoretical or true richness.

Moreno and Halfpeter (2000) suggest that the models should be built to compare different areas. In our case, the study was conducted along the same river, in locations where habitats differed only slightly and whose communities were consequently not too distinct. The good fit of CM to these data showed its reliability and its usefulness. The values of the coefficient of determination (R^2) were very high, exceeding the values obtained in a study conducted by Thompson et al. (2003) with Australian reptiles.

In The River Múrtigas, results for fifteen TACs, adjusted with CM, showed that on average about 15% of taxa richness per site remained undetected. Thus, the macroinvertebrate richness was insufficiently sampled for the purposes of ecological status indicators that require faunal composition. In fact, the 90% minimum for an acceptable inventory was never achieved (Gilbert et al., 2015). For example, according to CM, the true richness estimate for location 1 was 40 macroinvertebrate taxa. However, if faunal composition were necessary, only 35 taxa (88.5%) were actually captured and to achieve 90% five additional kicks would be needed, which would increase richness by just one taxon, still somewhat short of the estimated total.

Bradey and Ormerod (2002) extracted 70% of the total taxa richness in the first 5 samples, and Bady et al. (2005) obtained 50% in the first 10 samples. In this study, 10 kicks provided about 70% of total richness, except in four locations (3, 4, <60%; 13, 15, <70%). Therefore, the structural characteristics of macroinvertebrate communities along with the methodology employed enable the capture of a high proportion of richness in the first sampling units, although a much higher effort would be required to catch additional taxa.

It is not possible to give any kind of general recipe for a rapid bioassessment protocol to estimate the true richness based on a low fixed number of samples, because the reliability of the observed rich-

ness is dependent both on the curve shape (b) and the slope of the CM (a). The minimum number of samples needed to get an estimate of the true richness increases as the slope diminishes. On the other hand, the higher the slope, the higher the final percentage of total actual richness sampled ($R^2 = 0.89$, $p < 0.01$), which means that a lower number of samples will be needed to reach an acceptable estimate. The sample size must vary according to the initial diversity (Vlek et al., 2006). In studies of bioassessment where diversity and richness are high, a rapid assessment with a small sample size is inadequate to estimate the true richness (Thompson et al., 2003). Nevertheless, we tested this option by adjusting the CM to a sample size smaller than 20 kicks (3, 4, 5 and 10 kicks) to see whether it was possible to predict the true taxa richness with a minimum reliability with respect to the 20 kicks taxa accumulation curve asymptote. The predictions were highly irregular. Only 30%, 20%, 40% and 60% of the predictions for 3, 4, 5, and 10 kicks, respectively, were within the range of 90–110% of the actual 20 kicks predicted richness. From this it follows that it is not possible to recommend a rapid bioassessment based on less than the standard 20 kicks protocol.

4.3. Variability, accuracy and sample size

The errors in macroinvertebrate true richness estimates were dependent on sample size. Bias, precision and accuracy improved with sample size, following a specific tendency for each locality, more or less parallel but with different origins and ends. This means that comparisons between locations with the same effort can be made, but not between those with different levels of effort.

The use of TACs has been questioned in favour of non-parametric estimators, as they need large sample sizes to reach asymptotes and are sensitive to rare taxa (Thompson et al., 2003; Hortal et al., 2006). When the proportion of rare taxa is high, achieving a plateau in taxa richness found requires a much higher effort. This problem is accentuated when communities are patchily distributed, as is reflected in the shape of the sampling effort–richness curve (Gotelli and Colwell, 2001; Thompson and Withers, 2003; Willie et al., 2012). The use of non-parametric estimators with a relatively low sampling effort could be an alternative to the methodology employed here (e.g. Foggo et al., 2003), although, they can be similarly deficient with low sample sizes (Gotelli and Colwell, 2011). The reliability of non-parametric richness estimators with low sampling effort (e.g. Chao2, Bootstrap, Jackknife, ICE, etc.) depends on the TAC shape, i.e. the initial slope of the TAC needs to be steep to be reliable, as it means that the taxa are being quickly detected. Thus, the performance of these estimators ultimately also tends to depend on sampling effort, and it becomes necessary to take a large sample size to achieve a low biased and precise richness estimate (Walther and Moore, 2005; Hortal et al., 2006).

When the sampling effort has been exhaustive, observed richness is more reliable than non-parametric estimators (González-Oreja et al., 2010; Walther and Morand, 1998). If we compare our observed taxa richness with simulated data by Reese et al. (2014), our survey was more efficient in terms of error (S_{obs} : SME, -0.56 ; SMSE, 0.38 . See Appendix B). Our observed TAC exceeded the results obtained by the non-parametric estimators used in their study. Another study by Foggo et al. (2003) on invertebrate communities showed that their observed TACs were more accurate at initial sampling, with less erratic performance and less variance in their estimates, compared to non-parametric estimators (ICE, Chao 1, Chao 2), except Jackknife. The estimators improved with the sampling effort.

In our study, the ranks of estimated precision varied both across locations and levels of effort. As the sampling effort increased, the variance decreased to zero, therefore maximising precision. The loca-

tion with the highest values in the CV had lowest observed and predicted richness throughout the study (location 5). These results suggest that TAC is influenced by internal features of the community, which affects negatively the repeatability of sampling in the first units of effort, due to the large differences in richness extracted from permuted replicas. Successive units of effort (up to 20), however, allow the capture of a high percentage of actual richness (87.6%). If we compare it with the results obtained at location 4, also with a high CV, the number of taxa observed is much higher and the predicted richness reaches a lower final percentage (73.7%).

4.4. Implications and applications for water masses ecological status determination

The application of a systematic sampling protocol based on 20 kicks at 15 locations in a Mediterranean stream, both spatially and temporally homogeneous, proved to be statistically reliable in the estimation of true macroinvertebrate community richness through the CM with its corresponding error stated. However, it was insufficient to determine the current community composition, as on average five taxa were missed per location (range 2–10) from the final observed richness obtained after 20 kicks. Consequently, this study has demonstrated that the 20 kicks effort was not sufficient to reach an acceptable objective of 90% actual total richness. In fact, it remained on average as many as 18 kicks per location below the sample size that should be applied to achieve the aforementioned 90% objective (range 5–51 kicks).

Ideally, this sampling protocol needs to be cost-effective by achieving the maximum reliability with the minimum number of samples. However the possibility of a more or less rapid bioassessment based, for instance, on only 3, 4, 5 or 10 kicks has to be discarded because in the best option –10 kicks– only 60% of cases were in the range of 90–110% of the true richness estimates obtained with 20 kicks. The remaining cases were consistently below 50% (Appendix C).

Our results have shown that the minimum sample size, i.e. an acceptable objective of 90% community richness (Moreno and Halfpeter, 2000), was variable and location specific, probably conditioned by local habitat features and local community structure. In spite of the apparent temporally and spatially homogeneous ecological conditions registered in the River Múrtigas, the variability observed in richness estimates reliability and in the corresponding error statement prevent

the possibility of any general systematic proposal for adjusting the sample size in advance.

According to the results presented here, the inventory of actual taxa in a location has a variable level of uncertainty, which depends on the sampling effort. Therefore an index such as the BMWP could achieve a different quality category depending on the samples taken (Fig. 5). At locality 4, the maximum ecological quality –“very good”– was reached after 17 kicks and at locality 9, the highest quality “good” was reached after 8 kicks. Once the maximum quality, i.e. “very good”, was reached before 20 kicks, there were no further complications from a qualitative point of view, as any new taxa discovered would not improve the quality category, as in site 4. But what of locality 9? Here, the highest value attained was “good” and we know that there were still three undetected taxa. Considering all taxa found in both localities adjacent to 9, i.e. 8 and 10, that were not captured and were also abundant and well represented in the whole area of study, the three probable “lost” taxa could be Ephemerellidae, Philopotamidae and Siphonuridae. If we had taken enough kicks and captured them, the final category for locality 9 would have been “very good” instead of simply “good”. It follows that the establishment of the ecological status of any body of water will always be as arbitrary as the uncertainty of the total actual taxa richness, which, at the same time, will depend on the sampling effort. This is scientifically unacceptable.

Here we show that the sampling effort required to establish the full taxonomic composition of a macroinvertebrate community (the only scientifically acceptable manner of calculating qualitative indices such as BMWP) can entail unaffordable costs. Unfeasible budgets would be needed to determine the true richness (100% of the taxa present) as the effort to achieve such an objective can be higher than 100 kicks (Table 1). In this study there are two main points: 1) the estimation of true taxa richness can be reasonably accomplished with a 20 kicks sampling effort, and 2) the determination of the full taxonomic composition (to at least 90% richness), could require a much higher sampling to reliably calculate quality indices as the BMWP. Moreno and Halfpeter (2000) suggest 90% of the total estimated richness as the best cut-off point for a reliable unit effort, which would not take into account the hard-to-capture taxa, but would provide a large proportion of all the taxa present for most locations (Moreno and Halfpeter (2000); Willott, 2001; Thompson et al., 2007). The 90% threshold was proposed for comparing species richness between low gamma diversity communities made up of 20 bat

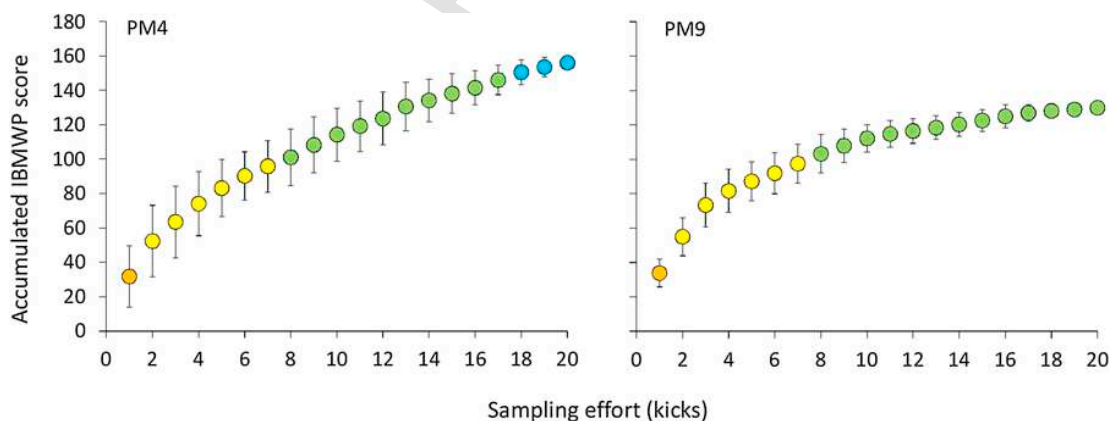


Fig. 5. The ecological status, according to BMWP scores based on macroinvertebrates, in Múrtigas river (SW Spain) depends on sampling effort. The locations with the most and less precise richness estimate, respectively (9 and 4) are represented. The curves have been obtained after 30 random permutations of the observed taxonomic composition, per kick, for each locality. Colour of circle represents quality states: Blue (Very Good); Green (Good); Yellow (Moderate); Orange (Poor). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

species. This cut-off point could be reduced in the case of more diverse communities, such as those of macroinvertebrate, thus reducing the sampling effort, but always in the knowledge that this reduction will have a negative non-stated impact on the final scores achieved in ecological quality indices, such as the BMWP. There are many other means of reducing the sampling effort and the global uncertainty associated with richness estimates. In studies on arachnids (Sørensen et al., 2002), it is recommended that studies focus on a few taxa, and that specific protocols be developed to minimize the problem of rare taxa. Other possibilities rely on the use of diversity indices or functional characteristics, which could provide greater accuracy than the direct use of taxa richness to describe the community, using a small number of samples or low sampling effort (Gayraud et al., 2003; Bady et al., 2005).

5. Conclusion

Specifying the reliability of richness estimates by stating the degree of error and determining the causes of uncertainty in the methodology, is a necessary step to establishing scientifically acceptable, quality, biological monitoring. This study shows that by using the 20 kicks methodology widely employed in WFD bioassessment, it is possible to achieve true macroinvertebrate richness estimates through the Clench Model, but the establishment of the community composition, i.e. the full taxa making up any index score, will be inaccurate to an unknown degree. Ultimately, this means that the ecological status assigned to a water mass could be somewhat arbitrary, to a degree that is proportional to the initial sampling decisions.

Uncited references

Clarke et al. (2006), Lande et al. (2000), Lopez et al. (2012)

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Appendix A. Abundance (number of individuals captured in 15 kicks) of macroinvertebrate families (except Hydrachnidae, Ostracoda and Oligochaeta) present in 15 locations studied at the Múrtigas river (SW Spain). *N*: sum of individuals in 15 sampling locations.

TAXA	1	2	3	4	5	6	7	8	9	10
ARACHNIDS										
Hydrachnidae				1						
COLEOPTERA										
Curculionidae										
Dytiscidae	8	4							1	
Elmidae	6	4	8	13	3	1	26	14	15	71
Gyrinidae	9	1								
Halipidae		1								
Hydrophilidae						7				
Hvgerbiidae				1						

Noteridae				1	3		2	2			
CRUSTACEANS											
Atyidae					1	3	18	176	48		
Ostracoda		43									
DIPTERA											
Athericidae		31	212	3	99	15	24	17		1	
Ceratopogonidae								3			
Chironomidae		295	298	37	127	55	90	215	108	24	
Culicidae			1								
Dixidae			1								
Limoniidae		6		2	1	20	24	34	75	82	
Psychodidae			6	1					1	1	
Rhagionidae		1									
Simuliidae		224	142		45		153	188	51	44	
Tabanidae					12		2	6	2	9	
Thaumaleidae											
Tipulidae		3	5	3							
EPHEMEROPTERA											
Baetidae		408	464	18	41	4	72	308	329	65	
Caenidae		38	77	5	94	40	45	192	189	54	
Ephemerellidae									38		
Heptageniidae		184	3								
Leptophlebiidae		11							2	2	4
Siphonuridae									3		
HETEROPTERA											
Corixidae					12	14	48	18	40	78	
Gerridae		7					7	167	105	7	
Mesoveliidae											
Naucoridae							1		2	52	
Pleidae										15	
HIRUDINEA											
Erpobdellidae		1	9	2							
Glossiphoniidae		34	36		2						
NEUROPTERA											
Sialidae			3		3						
MOLLUSCS											
Ancylidae		18	11		1						
Bithyniidae			48	1							
Ferrissidae											
Lymnaeidae		4	1								
Physidae			1						3	24	16
Planorbidae		47	1								
Sphaeriidae		11									
Unionidae								1			
ODONATA											
Aeshnidae			1	2	8	1					
Calopterygidae		3	12								
Coenagrionidae								12	39	12	
Cordulegasteridae		13	31	8	2						
Corduliidae										1	
Gomphidae		4	16	1	141	22	29	65	11	74	
Libellulidae					3				1	6	
Platycnemididae					1	1		3	1		
OLIGOCHAETA		142	152	2	13	22	136	131	28	48	
PLECOPTERA											
Capniidae		2									
Leuctridae		363	6		1						
Nemouridae		1									
TRICHOPTERA											
Calamoceratidae							3				
Ecnomidae								1			
Hydropsychidae		69	214	18	55		51	88	45	14	
Leptoceridae		1	3								
Philopotamidae			1								
Polycentropodidae		4									
Psychomyiidae		1		1							
Rhyacophilidae		11	9	4	2		1	1			
Sericostomatidae		27	9	1	1						
TURBELLARIA											
Dugesidae		48	3	2	2						
N		2078	1786	119	693	365	794	1384	1245	24	

Appendix B. Bias (scaled mean error, SME), precision (coefficient of variation, CV) and accuracy (scaled mean squared error, SMSE) of macroinvertebrates smoothed taxa accumulation curves (S_{est}) (100 permutations) for the 15 localities studied in the Múrtigas river (SW Spain), for five sampling efforts (3, 5, 10, 15 and 20 kicks). Calculations were performed on the asymptote predicted by Clench model.

		Locations						
	Kicks	1	2	3	4	5	6	7
Bias (SME)	3	-0.485	-0.577	-0.699	-0.705	-0.559	-0.558	-0.55
	5	-0.368	-0.460	-0.592	-0.605	-0.424	-0.443	-0.44
	10	-0.237	-0.297	-0.424	-0.462	-0.278	-0.287	-0.30
	15	-0.165	-0.203	-0.317	-0.347	-0.206	-0.208	-0.19
	20	-0.115	-0.146	-0.257	-0.263	-0.124	-0.164	-0.12
Precision (CV)	3	17.743	23.574	22.274	25.923	35.106	25.197	14.65
	5	12.053	15.587	16.398	16.241	25.300	16.702	11.35
	10	7.004	7.943	8.792	10.792	13.741	8.412	8.042
	15	4.227	3.490	5.397	7.481	10.910	4.553	6.041
	20	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Accuracy (SMSE)	3	0.243	0.343	0.494	0.503	0.336	0.324	0.309
	5	0.141	0.219	0.355	0.370	0.201	0.205	0.205
	10	0.059	0.092	0.183	0.217	0.087	0.086	0.093
	15	0.028	0.042	0.102	0.123	0.050	0.045	0.042
	20	0.013	0.021	0.066	0.069	0.015	0.027	0.015

Appendix C. Asymptote estimated at different sampling efforts through Clench Model and percentage respect asymptote estimated at 20 kick for sites sampled.

Sites	A from Kick 3		A from Kick 4		A from Kick 5		A from Kick 10		Kick 20
	%	%	%	%	%	%	%	%	
1	42.9	108.6	60.8	153.9	45.5	115.2	41.5	105.1	39.5
2	22.6	56.8	40.1	100.8	41.8	105.0	39.5	99.2	39.8
3	11.1	43.3	12.4	48.6	13.5	52.7	19.1	74.8	25.6
4	37.6	95.7	34.4	87.5	41.0	104.3	40.9	104.1	39.3
5	12.4	83.8	10.6	71.6	9.4	63.5	9.2	62.2	14.8
6	27.4	135.0	48.7	239.9	22.1	108.9	24.6	121.2	20.3
7	10.3	33.4	14.8	47.9	14.2	46.1	26.6	86.3	30.8
8	37.9	123.5	27.5	89.6	24.2	78.8	25.7	83.7	30.7
9	18.3	63.5	17.6	61.1	24.5	85.1	26.9	93.4	28.8
10	21.4	79.9	21.6	80.6	22.3	83.2	25.1	93.7	26.8
11	12.5	46.6	13.5	50.2	16.3	60.7	20.8	77.6	26.9
12	26.4	98.5	21.2	79.1	24.9	92.9	26.8	100.0	26.8
13	15.9	54.3	18.1	62.0	15.8	54.1	29.3	100.0	29.2

14	19.3	65.1	25.4	86.0	24.0	81.1	29.7	100.6	29.6
15	23.6	66.1	24.4	68.3	20.8	58.3	28.5	79.8	35.7
Av-er-age	22.6	76.9	26.1	88.5	24.0	79.3	27.6	92.1	29.6

Appendix D. Supplementary data

Supplementary data associated with this article can be found, in the online version, at <http://dx.doi.org/10.1016/j.ecolind.2016.07.047>.

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